The Feedback Vertex Set Problem: Theory, Algorithms, and Applications

CSE472 Project Presentation - Group 06

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Outline

1. Problem Definition and Visual Intuition
2. Complexity and NP-Completeness
3. Existing Algorithms Survey
4. Our implementations
5. Our Modifications
   - KMA: Kernelized Memetic Algorithm
   - DKMA: Dynamic Kernelized Memetic Algorithm
   - GNN-KMA Models
6. Results
   - Synthetic Dataset
   - PACE 2022 Benchmark
   - Results & Evaluation
7. Conclusions

Overview

Goal of This Section

- Understand what the Feedback Vertex Set (FVS) problem is
- Build intuition using cycles and forests
- Visualize how removing vertices eliminates cycles

What is a Cycle?

Definition

A **cycle** is a path that starts and ends at the same vertex without repeating edges.

- Cycles represent loops in a graph
- They create dependencies or redundancies

Key Idea

If a graph has no cycles, it is called a **forest**.

What is a Forest?

Definition

A **forest** is a graph with no cycles (i.e., a collection of trees).

- Trees are connected acyclic graphs
- Forest = one or multiple disconnected trees

Observation

Our goal in FVS is to transform any graph into a forest.

Feedback Vertex Set (FVS)

Definition

Given a graph $G = (V, E)$, a **Feedback Vertex Set** is a subset $S \subseteq V$ such that removing S makes the graph acyclic.

Objective

Find the **smallest possible** such set S.

- Remove vertices $\rightarrow$ eliminate cycles
- Minimize number of removed vertices

Problem Formulation

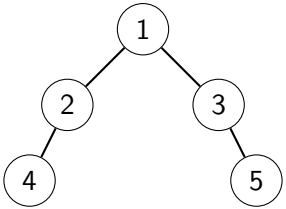
Decision Version

- Input: Graph $G = (V, E)$, integer k
- Question: Does there exist $S \subseteq V$ with $|S| \leq k$ such that $G - S$ is acyclic?

Optimization Version

Find a minimum-size Feedback Vertex Set.

Visual Example: Tree (No Cycles)

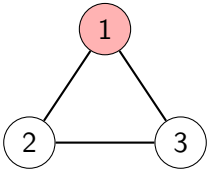


- No cycles exist
- Graph is already a forest

Result

FVS $= \emptyset$

Visual Example: Simple Cycle

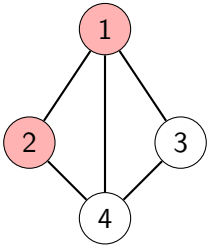


- This graph contains a cycle (triangle)
- Removing vertex 1 breaks the cycle

Result

FVS $= \{1\}$, size $= 1$

Visual Example: Multiple Cycles



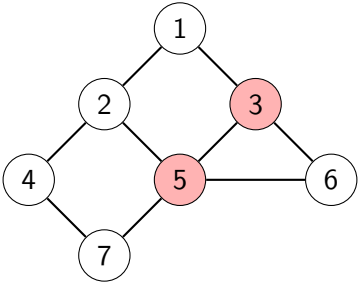
- Multiple overlapping cycles exist
- Need to remove more vertices

Result

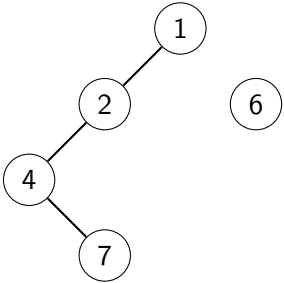
FVS $= \{1, 2\}$

Visual Example: Complex Graph → Forest

Before: Cycles



After: Forest



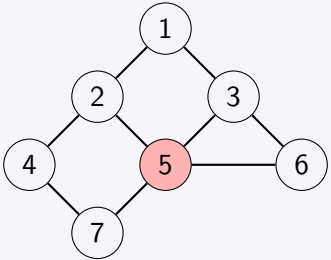
- **Before:** Multiple cycles exist, highlighted red vertices (3,5) are part of FVS.
- **After:** Removing FVS nodes breaks all cycles → Graph becomes a acyclic.

Final Verdict

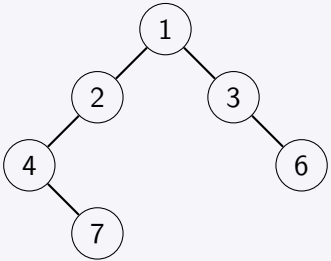
FVS $= \{3, 5\}$ → A set of vertices to remove all cycles.

Visual Example: Complex Graph → Minimal FVS → Forest

Original Graph with Cycles



Resulting Forest (Remove 5)



- Minimal Feedback Vertex Set: $\{5\}$
- Removing vertex 5 breaks all cycles → graph becomes a forest

Key Insight

Even if multiple vertices can break cycles, the **minimal FVS** is the set with the smallest size.

Real-World Motivation

- **Deadlocks in Operating Systems:** Cycles in resource allocation graph $\rightarrow$ remove critical process
- **Dependency Resolution:** Packages/modules with circular dependencies $\rightarrow$ remove one to break cycle
- **Circuit Design / Feedback Loops:** Loops in circuits $\rightarrow$ remove components to stabilize

Insight

FVS is a tool to identify minimal critical nodes whose removal resolves all cycles.

Key Observations

- Every cycle must contain at least one removed vertex
- Removing fewer vertices is always better
- Problem becomes harder as graph becomes denser

Core Insight

FVS is about selecting a minimum set of vertices that "hit" all cycles.

Complexity Overview

Goal of This Section

- Analyze the **computational complexity** of FVS
- Understand decision vs optimization versions
- Prove NP-Completeness using reductions

Decision vs Optimization

Decision Version

Given G, k: Does there exist an FVS of size $\leq k$?

Fact

Decision version is **NP-Complete**

Optimization Version

Find the minimum-size FVS

Fact

Optimization version is **NP-Hard**

Main Claim

Theorem

The decision version of Feedback Vertex Set is **NP-Complete**.

Proof Strategy

1. Show FVS $\in$ NP
2. Reduce Vertex Cover $\leq_p$ FVS

Step 1: FVS is in NP

Key Idea (NP Definition)

A problem is in NP if a solution can be **verified in polynomial time**.

- Given a set S
- Remove S from graph
- Check if graph has cycles (DFS)

Time Complexity

$O(V + E)$

Conclusion

FVS $\in$ NP (verification is polynomial)

Step 2: Reduction (VC $\rightarrow$ FVS)

Known Fact

Vertex Cover is NP-Complete

Goal

Transform VC instance into FVS instance

- Each edge becomes a cycle
- FVS must break all cycles

Step 2: Reduction (VC $\rightarrow$ FVS)

Vertex Cover (VC) Problem

Find smallest $S \subseteq V$ such that every edge has at least one endpoint in S.

Construction: Given VC instance (G, k)

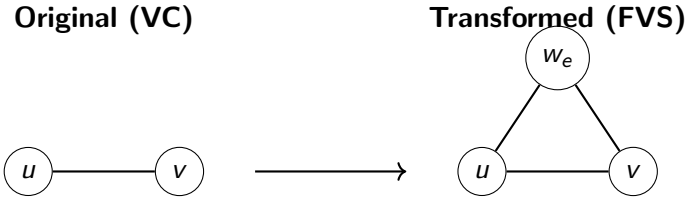
1. For each edge $e = (u, v) \in E(G)$:
   - Add new vertex w_e
   - Form triangle: edges (u, v), (u, w_e), (v, w_e)
2. Keep same parameter $k' = k$

Theorem

G has VC of size $\leq k \Leftrightarrow G'$ has FVS of size $\leq k$

Time: $O(|E|)$ (polynomial)

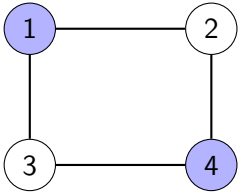
Reduction Construction (Graph)



- Each edge becomes a triangle
- Triangle = cycle → must be broken

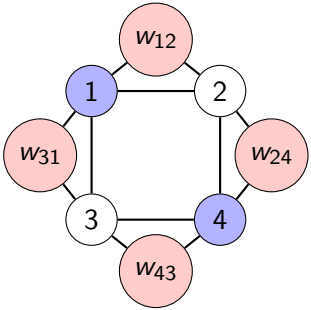
VC $\rightarrow$ FVS: Full Graph Illustration

Original Graph



VC $= \{1, 4\}$

Transformed Graph



Each triangle = one edge

Key Insight

Breaking all triangles = covering all edges

Correctness of Reduction

Claim

G has a VC of size $\leq k \iff G'$ has an FVS of size $\leq k$

Forward:

- VC selects endpoints $\rightarrow$ every triangle broken

Backward:

- FVS must hit each triangle
- Implies every edge is covered

Polynomial Time Reduction

- Add one vertex per edge
- Linear size increase

Conclusion

Reduction runs in polynomial time

Reverse Reduction (FVS $\rightarrow$ VC)

Reverse Direction

Shows Vertex Cover and FVS are polynomially equivalent

Construction: Given FVS instance (G, k)

1. Replace each vertex $v \in V$ with triangle $\{v_1, v_2, v_3\}$
2. For each edge $(u, v) \in E$, connect triangles with cross edges
3. Set $k' = 2|V| + k$

Theorem

G has FVS of size $\leq k \Leftrightarrow G'$ has VC of size $\leq 2|V| + k$

Key Insight:

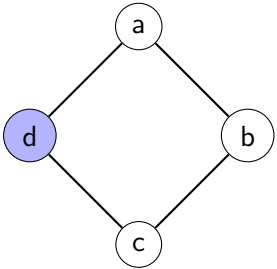
- Each triangle needs ≥ 2 vertices in VC
- Triangles with 3 vertices in VC, $\leftrightarrow$ vertices in FVS

FVS→ VC: Visual Example

Idea

Show structural similarity between problems

Original Graph



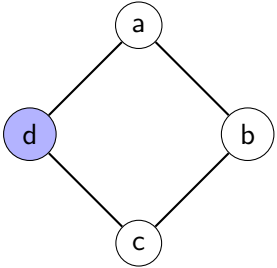
Construction

- Each vertex $\rightarrow$ triangle
- Selecting all 3, $\leftrightarrow$ remove vertex
- Selecting 2, $\leftrightarrow$ keep vertex

FVS $= \{d\}$

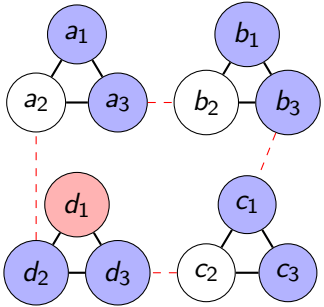
FVS→ VC: Visual Example

Original Graph G:



FVS: $\{d\}$ (size=1)
Removing d breaks the 4-cycle

Transformed Graph G':



Dashed = original edges , Each triangle = one vertex in G

Vertex Cover in G':(with d1) is a FVS in G'

Final Conclusion

We Have Proven

- FVS $\in$ NP
- Vertex Cover $\leq_p$ FVS
- VC is NP-Complete

Final Verdict

Feedback Vertex Set (Decision) is NP-Complete

Implication

- Optimization version is NP-Hard
- No known polynomial-time exact algorithm

Approaches to Solving FVS: Algorithmic Survey

Major Categories of Solution Approaches

- Exact Exponential Algorithms
- Approximation Algorithms
- Randomized Algorithms
- Heuristic Methods
- Metaheuristic Techniques

Objective

Provide a comprehensive overview of existing algorithms, their complexity, and practical trade-offs.

Exact Exponential Algorithms

Idea

Guarantee optimal solutions using exhaustive search or parameterized techniques.

Use Cases

- When exact solution is required
- Small graphs or small parameter k

Algorithms

- Naive Brute Force
- DP on Subsets
- Iterative Compression
- Bounded Search Tree
- Current Best FPT
- Inclusion-Exclusion DP

Naive Brute Force

Algorithm Steps

- Generate all subsets of vertices
- For each subset, remove vertices from graph
- Check if resulting graph is acyclic
- Return smallest valid subset

Complexity

$O(2^n \cdot n^3)$

Reference

Garey & Johnson (1979)

DP on Subsets

Algorithm Steps

- Enumerate subsets of vertices
- Store intermediate results using DP
- Avoid recomputation of overlapping subsets
- Select minimum valid FVS

Complexity

$O(2^n \cdot n^2)$

Reference

Cygan et al., Parameterized Algorithms (2015)

Iterative Compression

Algorithm Steps

- Start with small subgraph solution
- Add vertices incrementally
- Maintain solution of size $k + 1$
- Compress solution to size k

Complexity

$O(5^k \cdot kn^2)$

Reference

Dehne et al., Theory of Computing Systems (2007)

Bounded Search Tree

Algorithm Steps

- Detect a cycle in the graph
- Select a vertex from the cycle
- Branch by removing each possible vertex
- Recurse until graph becomes acyclic or k exhausted

Complexity

$O(4^k \cdot n^2)$

Reference

Chen et al. (2008)

Current Best FPT

Algorithm Steps

- Apply reduction and kernelization rules
- Use advanced branching strategies
- Reduce problem size iteratively
- Solve using optimized search tree

Complexity

$O(3.619^k \cdot n^4)$

Reference

Chen et al., Journal of Computer and System Sciences (2008)

Inclusion-Exclusion DP

Algorithm Steps

- Enumerate subsets using inclusion-exclusion principle
- Count valid configurations using DP
- Avoid overcounting via inclusion-exclusion
- Select subset forming valid FVS

Complexity

$O(2^n \cdot n)$

Reference

Björklund et al. (2010)

Approximation Algorithms

Idea

Provide near-optimal solutions with provable approximation guarantees.

Use Cases

- When exact solution is computationally infeasible
- Large graphs where guarantees are required

Algorithms

- 2-Approximation (VC-based)
- Log-Approximation
- LP Rounding
- Primal-Dual

2-Approximation (VC-based)

Algorithm Steps

- Find a cycle in the graph
- Select an edge from the cycle
- Remove both endpoints of the edge
- Repeat until graph becomes acyclic

Approximation Guarantee

$\leq 2 \times OPT$

Reference

Bafna et al. (1999)

Log-Approximation

Algorithm Steps

- Convert problem into hitting set formulation
- Identify sets corresponding to cycles
- Apply greedy/log approximation algorithm
- Select vertices covering all cycles

Approximation Guarantee

$O(\log n)$

Reference

Bar-Yehuda (Approximation for FVS)

LP Rounding

Algorithm Steps

- Formulate FVS as integer linear program
- Relax constraints to linear program
- Solve LP to obtain fractional solution
- Round values to obtain integer solution

Complexity

$O(n^{3.5})$

Reference

Bar-Yehuda & Even (1985)

Primal-Dual

Algorithm Steps

- Initialize primal and dual solutions
- Increase dual variables iteratively
- Select vertices when constraints become tight
- Construct feasible FVS solution

Complexity

$O(n^2)$

Reference

Goemans & Williamson (1995)

Randomized Algorithms

Idea

Use randomness to achieve good expected performance and scalability.

Use Cases

- When probabilistic guarantees are acceptable
- Large-scale graphs with time constraints

Algorithms

- Random Sampling
- Randomized Rounding
- Color Coding

Random Sampling

Algorithm Steps

- Randomly select a vertex from graph
- Check if it belongs to a cycle
- Add it to solution if needed
- Repeat multiple times and choose best result

Performance

Expected $O(\log n)$ approximation

Reference

Motwani & Raghavan, Randomized Algorithms

Randomized Rounding

Algorithm Steps

- Solve LP relaxation
- Interpret fractional values as probabilities
- Randomly select vertices based on probability
- Repeat and select best solution

Guarantee

Expected 2-approximation

Reference

Vazirani (2001)

Color Coding

Algorithm Steps

- Assign random colors to vertices
- Identify colorful structures
- Use dynamic programming to detect solutions
- Repeat to improve probability

Complexity

$O(2^{O(k)} \cdot n)$

Reference

Alon, Yuster, Zwick (1995)

Heuristic Algorithms

Idea

Provide fast and practical solutions without theoretical guarantees.

Use Cases

- Real-world applications requiring fast solutions
- Very large graphs where exact methods fail

Algorithms

- Greedy Vertex Selection
- Cycle-Based Greedy
- Maximum Degree Heuristic
- Core-Periphery Decomposition
- Reduction Rules + Greedy

Greedy Vertex Selection

Algorithm Steps

- Compute degrees of all vertices
- Select highest-degree vertex
- Remove it from graph
- Repeat until no cycles remain

Complexity

$O(n^2 \log n)$

Reference

Festa et al., Theoretical Computer Science (2004)

Cycle-Based Greedy

Algorithm Steps

- Detect cycles using DFS/BFS
- Count vertex participation in cycles
- Select most frequent vertex
- Remove and repeat

Complexity

$O(n^3)$

Reference

Festa et al., European Journal of Operational Research (2008)

Maximum Degree Heuristic

Algorithm Steps

- Compute vertex degrees
- Apply tie-breaking rules
- Remove highest scoring vertex
- Repeat until acyclic

Complexity

$O(n^2 \log n)$

Reference

Fomin et al., Discrete Applied Mathematics (2006)

Core-Periphery Decomposition

Algorithm Steps

- Compute k-core decomposition
- Identify dense core region
- Focus removal on core vertices
- Apply greedy selection

Complexity

$O(n + m)$

Reference

Batagelj & Zaversnik, SIAM (2003)

Reduction Rules + Greedy

Algorithm Steps

- Remove degree ≤ 1 vertices
- Contract degree-2 vertices
- Apply preprocessing rules
- Run greedy on reduced graph

Complexity

$O(n^2)$

Reference

Kernelization: New Upper and Lower Bound Techniques

Metaheuristic Algorithms

Idea

Use global optimization techniques to explore large solution spaces.

Use Cases

- Complex large-scale graphs
- When near-optimal solutions are acceptable

Algorithms

- Genetic Algorithm
- Simulated Annealing
- Ant Colony Optimization
- Tabu Search
- Particle Swarm Optimization

Genetic Algorithm

Algorithm Steps

- Initialize population randomly
- Evaluate fitness of solutions
- Apply crossover and mutation
- Select best individuals iteratively

Complexity

$O(g \cdot p \cdot n^2)$

Reference

Goldberg (1989)

Simulated Annealing

Algorithm Steps

- Start with initial solution
- Generate neighboring solution
- Accept worse solutions probabilistically
- Gradually decrease temperature

Complexity

$O(i \cdot n^2)$

Reference

Kirkpatrick et al. (1983)

Ant Colony Optimization

Algorithm Steps

- Initialize pheromone levels
- Construct solutions probabilistically
- Update pheromones based on quality
- Repeat for multiple iterations

Complexity

$O(\text{ants} \cdot i \cdot n^2)$

Reference

Dorigo & Gambardella (1997)

Tabu Search

Algorithm Steps

- Start with initial solution
- Explore neighborhood solutions
- Maintain tabu list to avoid repetition
- Update best solution found

Complexity

$O(i \cdot n^2)$

Reference

Glover (1989)

Particle Swarm Optimization

Algorithm Steps

- Initialize swarm of particles
- Update velocity and position
- Share global and local best
- Iterate until convergence

Complexity

$O(\text{particles} \cdot i \cdot n^2)$

Reference

Kennedy & Eberhart (1995)

Algorithm Selection Criteria

Essential Criteria

1. **Implementability**
   - Feasible in 6-week timeframe
   - Well-documented in literature
   - Reasonable debugging complexity
2. **Research Potential**
   - Novel comparisons possible
   - Room for improvements
   - Publishable results
3. **Complementary Nature**
   - Different problem sizes
   - Different paradigms
   - Interesting comparisons

Additional Criteria

4. **Practical Relevance**
   - Real-world applications
   - Industry interest
   - Addresses practical needs
5. **Educational Value**
   - Learn important techniques
   - Transferable skills
   - Deep algorithmic insights

Goal

Select algorithms that maximize research impact while ensuring feasibility

Selected Algorithms for Implementation

Three-Algorithm Approach (Maximum Research Impact)

We will implement three complementary algorithms covering the entire solution spectrum:

1. **Iterative Compression** (Exact FPT Algorithm)
   - Guarantees optimal solution for small-medium k
   - Time: $O(5^k \cdot kn^2)$ — FPT breakthrough (2004)
   - Research: Hybrid approaches, parallelization, adaptive branching
2. **Kernelization + Bounded Search Tree** (Enhanced Exact)
   - Preprocessing with reduction rules + exact search
   - Kernel reduction: 50-90% for many graph types
   - Research: New reduction rules, smart branching strategies
3. **Memetic Algorithm** (Advanced GA + Local Search)
   - Handles large instances (n > 1000)
   - Combines population evolution with local optimization
   - Research: Problem-specific operators, ML integration, hybrid exact-metaheuristic

Algorithm 1: Iterative Compression (IC)

Core Concept

Given FVS of size $k + 1$, compress it to size k through smart enumeration

Theoretical Significance:

- Most important FPT technique
- Published in STOC/FOCS
- Demonstrates advanced understanding

Implementation:

- Complexity: Medium
- Expected range: $k \leq 30$
- Validation on small instances

Research Directions:

1. **Hybrid Kernelization + IC**
   - Apply reduction rules before IC
   - Potential 2-5x speedup
2. **Adaptive Branching**
   - Learn which vertices to branch first
   - Use graph structure metrics
3. **Parallelization**
   - Independent branches in parallel
   - Multi-core speedup potential
4. **Improved Compression**
   - Smart partition enumeration
   - Structure-based pruning

Iterative Compression for FVS

Idea: Build solution incrementally and compress when too large.

Algorithm

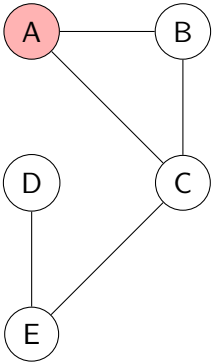
```
Order vertices v_1, ..., v_n
F ← {v_1}
for i = 2 to n do
    F ← F ∪ {v_i}
    if |F| = k + 1 then
        for all partitions (F_1, F_2) of F do
            find Y ⊆ V \ F_2
            if F_1 ∪ Y is an FVS then
                update F and break
            end if
        end for
        if no update performed then
            return NO
        end if
    end if
end for
return F if |F| ≤ k
```

Key Insight: Compress size $k+1$ to k

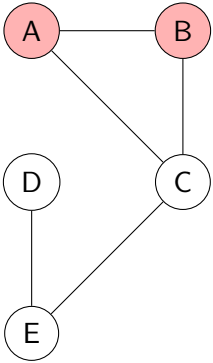
Time: $O(5^k \cdot n^2)$

Simulation

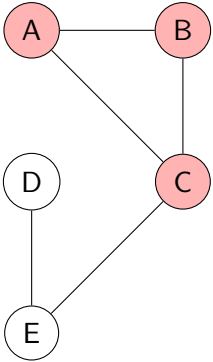
Simulation of FVS (k=2) on 5 vertices
Step 1: $F = \{A\}$



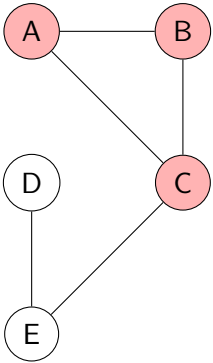
Step 2: $F = \{A, B\}$



Step 3: $F = \{A, B, C\}$ (overflow)

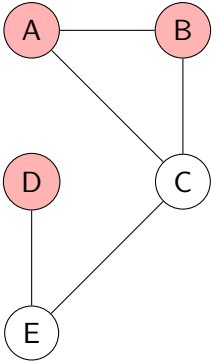


Step 3: $F = \{A, B, C\}$ (overflow)

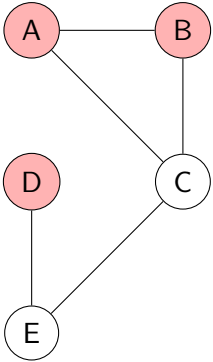


Fix: Remove C $\Rightarrow F = \{A, B\}$

Step 4: Add D (Overflow)

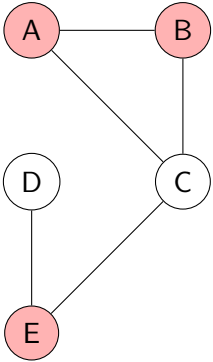


Step 4: Add D (Overflow)

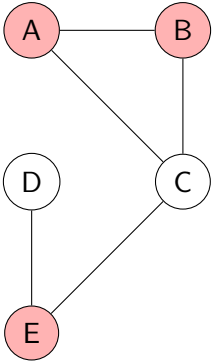


Fix: Remove D $\Rightarrow F = \{A, B\}$

Step 5: Add E (Overflow)

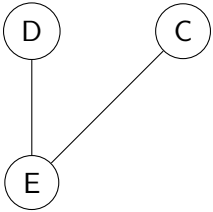


Step 5: Add E (Overflow)



Fix: Remove E $\Rightarrow F = \{A, B\}$

Final Answer



$F = \{A, B\}$

Algorithm 2: Kernelization + Bounded Search Tree

Core Concept

Reduce graph size via preprocessing rules, then apply exact search on smaller kernel

Reduction Rules:

- Degree 0, 1, 2 vertices
- Self-loops, duplicate edges
- Crown decomposition
- Flower rule

Search Strategy:

- Smart vertex selection
- Bounded depth: k
- Branching factor: ≤ 4

Research Directions:

1. **New Reduction Rules**
   - Graph-specific rules
   - Theoretical contribution
2. **Rule Application Order**
   - Optimize ordering
   - ML-based prediction
3. **Kernelization Bounds**
   - Tighter bounds for special classes
   - Experimental validation
4. **Smart Branching**
   - Centrality-based selection
   - ML-guided search

Kernelization + Bounded Search Tree

Idea: Reduce the graph, then branch on remaining cycles.

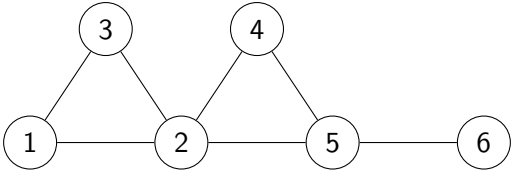
Algorithm

```
Apply reduction rules:
    remove degree ≤ 1 vertices
    contract degree-2 vertices
    include self-loop vertices
Obtain kernel (G', k')
function Search(G', k')
if G' is acyclic then
    return solution
end if
if k' < 0 then
    return NO
end if
find a cycle C
for each v ∈ C do
    recurse on (G' − v, k' − 1)
end for
```

Key Insight: Small kernel $\Rightarrow$ small search tree

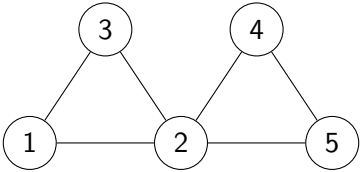
Time: $O(4^k + n^2)$

Step 1: Original Graph

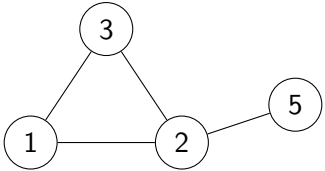


Step 2: Remove degree ≤ 1 vertices

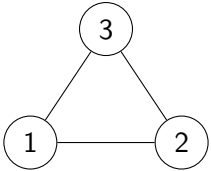
Vertex 6 removed



Step 3: Contract degree-2 vertex (4)

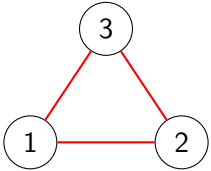


Step 4: Remove degree ≤ 1 vertex (5)

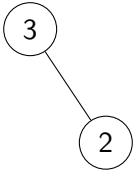


Step 5: Kernel Graph (G')

Remaining cycle

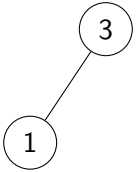


Step 6: Branch - Remove vertex 1



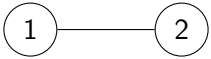
Graph becomes acyclic

Step 7: Branch - Remove vertex 2



Graph becomes acyclic

Step 8: Branch - Remove vertex 3



Graph becomes acyclic

Algorithm 3: Memetic Algorithm (GA + Local Search)

Core Concept

Population evolution + local optimization for large-scale instances

Components:

- Smart initialization (greedy + random)
- FVS-aware crossover operators
- Centrality-based mutation
- Problem-specific local search

Practical Relevance:

- Industry standard
- Handles $n > 1000$ vertices
- Near-optimal (5-10% gap)

Research Directions:

1. **Cycle-Preserving Crossover**
   - Preserve cycle-breaking structures
   - Novel contribution
2. **Hybrid Exact-Metaheuristic**
   - Use IC for small components
   - GA for large core
3. **ML Integration**
   - Predict promising vertices
   - Learn operator selection
4. **Multi-Objective**
   - Minimize size + maximize robustness
   - Pareto frontier

Memetic Algorithm for FVS

Idea: Evolutionary search combined with local optimization.

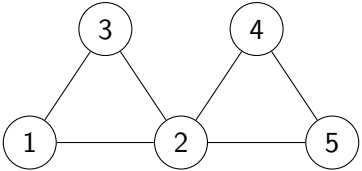
Algorithm

```
Encode solution as a permutation of vertices
Decode by removing vertices until graph is acyclic
Initialize population (greedy + random)
for each generation do
    Evaluate fitness: −|FVS|
    Select parents (tournament)
    Crossover (cycle-aware)
    Mutation (swap cycle-heavy vertices)
    Local search (hill-climbing)
end for
return best solution found
```

Key Insight: Exploration + refinement

Type: Heuristic (near-optimal)

Step 1: Original Graph

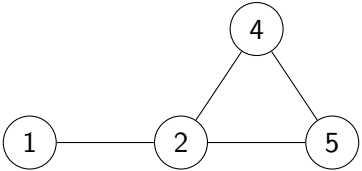


Step 2: Encode Solution

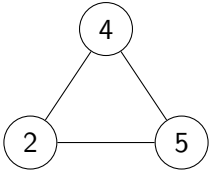
Example permutation:

$$[3, 1, 5, 2, 4]$$

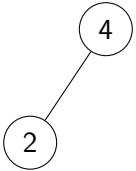
Step 3: Remove vertex 3



Step 4: Remove vertex 1



Step 5: Remove vertex 5



Graph is now acyclic $\rightarrow$ Stop decoding

Step 6: Fitness Evaluation

$$\text{Removed vertices} = \{3,1,5\}$$

$$\text{Fitness} = -|FVS| = -3$$

Step 7: Selection (Tournament)

Select best solutions from population

Example:

$$[3, 1, 5, 2, 4], \quad [2, 4, 1, 5, 3]$$

Step 8: Crossover

Parent 1: [3,1,5,2,4]
Parent 2: [2,4,1,5,3]

Child:

[3,4,1,2,5]

Step 9: Mutation

Before:

$$[3, 4, 1, 2, 5]$$

After swap:

$$[3,1,4,2,5]$$

Step 10: Local Search

Try improving solution:

Remove fewer vertices if possible

Example improvement:

$$\text{FVS size: } 3 \rightarrow 2$$

Step 11: Best Solution

Best FVS found:

$$\{3,2\}$$

Graph becomes acyclic

Our Modifications

KMA DKMA GNN-KMA

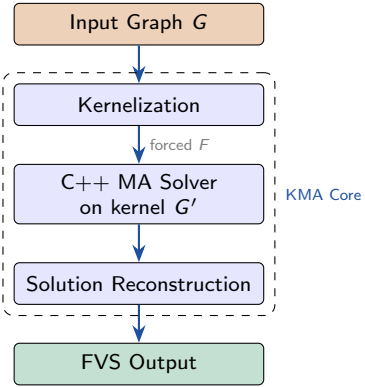
KMA - Motivation and High-Level Design

Why not pure Memetic Algorithm?

- A plain MA [7] operates on the full graph
- Reduction rules can *safely* eliminate many vertices before any search [3]
- This shrinks the search space, letting MA focus on hard subgraphs

KMA Pipeline

1. **Kernelization** - apply reduction rules, record forced vertices
2. **MA on kernel** - C++ evolutionary search on reduced graph
3. **Reconstruction** - map kernel solution to original indices, union with forced vertices



KMA - Kernelization Rules

Undirected Kernelization Rules

1. **Self-loop:** vertex with self-loop $\Rightarrow$ force into FVS
2. **Degree-0/1:** acyclic leaf $\Rightarrow$ safely delete
3. **Degree-2 bypass:** if $\deg(v) = 2$ with neighbors a, b and $\{a, b\} \notin E$: delete v, add edge (a, b)

Rules applied **iteratively** until quiescence.

Theoretical basis: $O(k^2)$ kernel [3].

Directed Kernelization Rules

1. **Self-loop:** forced into DFVS
2. **Source/Sink:** in-deg $= 0$ or out-deg $= 0 \Rightarrow$ delete (not on any directed cycle)
3. **Degree-1 bypass:** in-deg or out-deg $= 1 \Rightarrow$ shortcut edges, delete vertex
4. **SCC reduction:** retain only vertices in nontrivial SCCs (size > 1 or self-loop)

Directed kernel theory: [6].

Modification over baseline MA: The addition of kernelization (especially the SCC-based directed reduction) is a key novelty. SCC reduction alone can eliminate **60–90%** of vertices on sparse real-world directed graphs before any search begins.

KMA - Memetic Algorithm Core

C++ MA Solver (evolutionary search on kernel):

- Initialise population of size P randomly
- Each individual: binary vector over kernel vertices
- Each generation:
  1. **Crossover** (two parents $\rightarrow$ offspring)
  2. **Local mutation** (flip, swap)
  3. **Repair**: restore FVS validity via greedy cycle cover
  4. **Selection**: tournament
- Early-stop if no improvement for T generations
- Hard wall-clock timeout enforced inside C++

Parameters

Param	Flag	Default
Population	`--pop`	20
Max generations	`--gens`	100
Timeout	`--timeout`	600s
Early-stop	`--earlystop`	30

Complexity Note

Kernelization: $O(n + m)$ per pass
MA on kernel G': heuristic, $O(P \cdot \text{gen} \cdot |V(G')|)$
Full pipeline: practical for $|V| \leq 10^4$ within timeout

DKMA - Dynamic Kernelized Memetic Algorithm - Core Idea

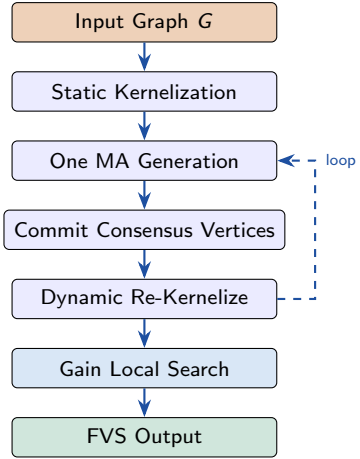
Limitation of static kernelization (KMA):

- Kernelization is applied *once* before MA
- Partial solutions during MA may *unlock* new reductions - these are never exploited

DKMA - Dynamic interleaving [8, 9]:

1. Evolve population for one MA generation
2. **Commit** vertices appearing in $\geq \tau$ fraction of individuals
3. Force committed vertices into the solution
4. **Re-kernelize** residual graph
5. Continue MA on the smaller kernel

This **opening-up effect** progressively shrinks the kernel as partial agreement is reached.



DKMA - Key Innovations & Helpers

SHORTONE Directed Contractions

Inspired by the Mount-Doom PACE 2022 solver [10], a directed contraction pass (`_apply_shortone_rule`) is applied before the MA loop on directed graphs.

Contracts degree-1 in/out vertices with special cycle-preserving rewiring, enabling additional SCC reductions.

Gain Local Search (post-loop)

`_gain_local_search` applies **1-1** and **2-1** improvement swaps after the MA loop:

- **1-1 swap**: remove one FVS vertex, add another if acyclicity is preserved and size decreases
- **2-1 swap**: replace two FVS vertices with one

Topological Diversification

Every 5 generations, `_diversify_with_topological_ordering` is called:

- Directed: randomised Kahn's algorithm tie-breaking
- Undirected: random spanning-tree cycle-breaking insertion

Prevents premature convergence [7].

Commit Threshold & Cap

`_commit_vertices_from_population`:

- Commit if vertex appears in $\geq \tau$ of population
- Cap at top 30% by frequency if over-committing
- Default $\tau = 0.8$

Based on DFVS IPEC 2022 approach [8].

DKMA - Correctness & Comparison with KMA

Correctness Safeguards

- Deduplicated final solution indices
- All forced vertices (static + dynamic) included
- Final **acyclicity verification** on original graph: directed $\Rightarrow$ iterative DFS back-edge detection;
  undirected $\Rightarrow$ union-find cycle check
- **Fallback to KMA** if validation fails

KMA vs DKMA - Key Differences

Feature	KMA	DKMA
Kernelization	Once	Per loop
Commitment	None	Consensus
SHORTONE pass	No	Yes (directed)
Topological div.	No	Every 5 gen.
Local search	No	Gain 1-1/2-1
Acyclicity check	End	End + fallback

References: DKMA design is inspired by the PACE 2022 FVS track winner strategies [8] (Grossmann, Heuer, Schulz, Strash, IPEC 2022) and the ESA 2022 DFVS analysis [9] (Figiel, Froese, Nichterlein, Niedermeier). The Mount-Doom SHORTONE rule is from Langedal et al. [10].

GNN-KMA - Hybrid Coupling Strategy

Motivation:

- GNN learns vertex-level patterns from solved instances
- Predicts probability $p(v)$ of belonging to FVS
- High-confidence predictions **hard-fix** vertices before KMA, reducing the search problem

Precision-first soft-hint coupling:

1. Kernelize graph $G \to G'$
2. Run GNN on G', get $p(v)$ for each v
3. Hard-fix only vertices with $p(v) \geq \theta$
4. Cap hard-fix at **8%** of kernel size
5. Run KMA on residual kernel
6. Reconstruct: forced $\cup$ GNN-fixed $\cup$ KMA output

Default: $\theta = 0.65$, min-fix $= 0.5\%$, max-fix $= 8\%$ of kernel

Why Precision-First?

False positives in hard-fixing are **catastrophic**: a wrongly fixed vertex inflates the FVS size permanently within that branch. Capping at 8% and requiring $\theta \geq 0.65$ ensures the GNN only commits when highly confident.

If no vertex exceeds θ, the system **falls back to pure KMA** automatically.

Can you do a GNN-DKMA variant? Same coupling, but the residual kernel is handed to DKMA instead of KMA, enabling dynamic re-kernelization atop GNN guidance.

GNN-KMA v1 - Baseline Architecture

Feature Engineering (3 channels)

Undirected:

- Normalised degree $d(v)/d_{\max}$
- Clustering coefficient $c(v)$ [18]
- Normalised log-degree

Directed:

- Normalised in-degree
- Normalised out-degree
- Normalised min(in, out)-degree

Design Choice

Clustering coefficient [18] captures local cycle density - vertices with high $c(v)$ are strong FVS candidates. This is a well-established heuristic from the small-world network literature.

Model Architecture

UndirectedFVSNet:

- 3-layer **GraphSAGE** message passing [11]
- BatchNorm + Dropout after each layer
- MLP classification head
- 2-class `log-softmax` output

DirectedFVSNet:

- Custom directed aggregation layer (separate in / out / self terms)
- 3 layers + BN + Dropout + MLP
- 2-class `log-softmax` output

Training Objective

Weighted NLL loss + cosine LR schedule.
Checkpoint by **validation F1** score.

GraphSAGE [11]: Hamilton et al., NeurIPS 2017.

GNN-KMA v2 - Richer Structural Features

Feature Engineering (11 channels)

1. Degree-like channels (in/out/min or undirected)
2. **RWSE** - Random Walk Structural Encoding [13] short-range return probabilities at steps $\{2, 3, 4\}$
3. **Motif counts**: triangle count, 4-cycle participation, 4-clique style counts [14]
4. **k-core number** $\kappa(v)$ [15]

Architecture & Training

UndirectedFVSNetV2 / DirectedFVSNetV2:

- Same GraphSAGE-style backbone as v1
- Upgraded input: `in_channels=11`
- Richer representation $\Rightarrow$ stronger structural discrimination

Training: same NLL + cosine schedule; checkpoint by validation F1.

Why These Features?

RWSE [13]: captures short-range cycle return probability - directly relevant to FVS membership.

k-core [15]: vertices deep in dense cores are more likely to be in any FVS.

Motif counts [14]: triangle/4-cycle participation indicates cycle exposure.

GNN-KMA v3 - Attention-Based Precision Model

Feature Engineering (16 channels)

- Extended **RWSE** horizon: steps $\{2, 3, 4, 6, 8, 12, 16\}$ [13]
- **SCC-aware channels** for directed cycle structure (SCC size, SCC membership depth)
- Cycle-score interaction features
- Directional ratio features (in/(in + out))
- Reduced reliance on expensive motif channels

SCC Channels (Directed)

SCC decomposition [16] provides structural depth: vertices in large SCCs are more critical for DFVS. SCC-size and SCC-depth channels directly encode this.

DirectedFVSNetV3 Architecture

- Input projection layer
- **5 residual GATv2 blocks** [12]
- Separate **forward** and **reverse** message paths
- Per-layer fusion projections
- **Global readout context** concatenated to node embedding
- Single-logit output per node (sigmoid)

v3 Training Innovations

- **AsymmetricFVSLoss**: penalises false positives more strongly (precision-oriented)
- Warmup + cosine LR + **gradient clipping**
- Checkpoint metric: **topk-precision@8%** (matches hard-fix cap)
- 200 epochs (vs 100 for v1/v2)

GNN-KMA - Feature & Architecture Evolution Summary

Aspect	**v1**	**v2**	**v3**
Feature channels	3	11	16
Degree features	✓	✓	✓
Clustering coeff. [18]	✓	–	–
RWSE [13]	–	steps $\{2, 3, 4\}$	steps $\{2..16\}$
Motif counts [14]	–	✓	reduced
k-core [15]	–	✓	–
SCC channels [16]	–	–	✓
Backbone	GraphSAGE [11]	GraphSAGE	GATv2 [12]
Depth	3 layers	3 layers	5 residual blocks
Bidirectional	No	No	Yes (fwd+rev)
Global readout	No	No	Yes
Loss	NLL	NLL	AsymmetricFVSLoss
Checkpoint metric	Val F1	Val F1	topk-prec@8%
Epochs	100	100	200

Results

Synthetic Dataset PACE 2022 Evaluation

Notation & Terms

- **MA**: Memetic Algorithm - a genetic algorithm with local search
- **KMA**: Kernelized Memetic Algorithm - MA applied on a reduced kernel
- **DKMA**: Dynamic Kernelized Memetic Algorithm - KMA with iterative re-kernelization
- **GNN-KMA**: GNN-augmented KMA - GNN guides/fixes high-confidence vertices for KMA

Dataset - Design Goals & Generation Pipeline

Five design goals:

1. **Controlled distribution** - fixed category proportions
2. **Track-aware difficulty** - separate exact & heuristic tracks
3. **Reproducibility** - deterministic seeding throughout
4. **Restartability** - safe reuse of existing files
5. **Benchmark/train alignment** - PT datasets mirror TXT corpus structure

Three-stage orchestration:

1. `download_real_world.py`
   NetworkX, PyG, OGB, SNAP sources
2. `generate_synthetic.py`
   Barabási–Albert, Watts–Strogatz, ER, grids, trees, DAGs
3. `dataset_gen.py`
   TXT → PyG Data objects with solver labels

Directory Contract

```
data/synthetic/
{undirected|directed}/
{exact_track|heuristic_track}/
{category}/
*.txt
```

Track Size Intent

Track	Node range
exact_track	≈ 10–35 nodes
heuristic_track	≈ 100–5000 nodes

Default: 100K undirected + 100K directed
exact ratio = 0.5, seed = 1337

Dataset - Undirected Distribution

Undirected Categories (20% each)

Category	Weight	Graph Model
`real_world`	20%	NetworkX/PyG/OGB/SNAP
`scale_free`	20%	Barabási–Albert [17]
`small_world`	20%	Watts–Strogatz [18]
`random_er`	20%	Erdős–Rényi [19]
`grids_trees`	20%	Grid + Random tree

Note: The undirected dataset balances real-world graphs, scale-free networks, small-world models, random Erdős–Rényi graphs, and structured grids/trees.

Dataset - Directed Distribution

Directed Categories

Category	Weight
`real_world_ego`	30%
`scale_free`	20%
`random_er`	20%
`directed_grids`	15%
`dags`	15%

Note: Directed instances include ego-networks, scale-free digraphs, random directed graphs, grid-based graphs, and acyclic DAG structures.

Dataset - Real-World Share & Labels

Real-world share:
Undirected: 20% real-world graphs
Directed: 30% ego-network graphs

Sources: NetworkX built-ins, PyG datasets, OGB node property datasets, SNAP archives.

Cached under `.cache/real_graphs`.

Label Sources

exact_track: IC solver labels
(high-fidelity, near-optimal)

heuristic_track: KMA labels
(practical, large-instance behavior)

Dataset - Why We Used This Approach

Diversity

- **5 undirected families**: covers scale-free hubs, small-world clusters, random structure, regular grids/trees, and real topology
- **5 directed families**: adds ego networks and DAGs
- Both undirected and directed instances
- Two difficulty regimes (exact / heuristic)

Quality Guarantees

- Each label validated via acyclicity check
- Subprocess timeout isolation for all solver calls
- CSV status ledger tracks completed / timeout / invalid outcomes
- Largest connected component extracted
- Compact integer relabelling applied
- No self-loops in edge lists

Scale

100K total undirected + 100K directed
(20K version also available on Kaggle)

Largest publicly available FVS dataset with ground-truth labels to our knowledge.

Kaggle (100K):
`kaggle.com/.../fvs-synthetic-dataset-100k`

Kaggle (20K):
`kaggle.com/.../fvs-synthetic-dataset-20k`

PACE 2022 Dataset - Real-World Benchmark Testing

What is PACE 2022?

The **Parameterized Algorithms and Computational Experiments** (PACE) Challenge 2022 [8] focused on the **Feedback Vertex Set** problem.

It provided:

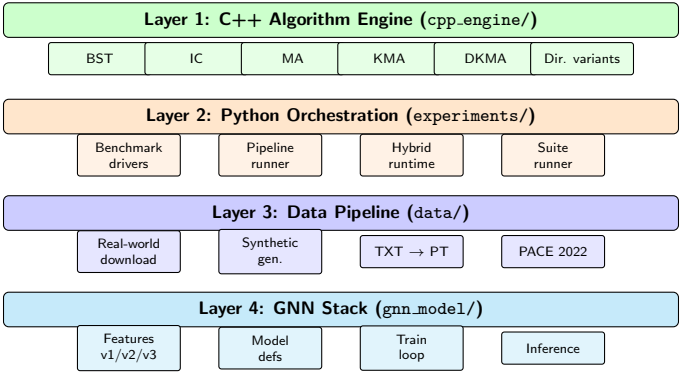
- 200 public undirected test instances
- Curated from real-world and synthetic sources
- Difficulty ranges from small exact to large heuristic
- Official solution checker for validation
- World-class solver implementations as baselines

Usage in This Project

- PACE 2022 instances stored under `data/pace2022/`
- **Protected from cleanup** - never deleted by `setup_benchmark_inputs.py`
- Benchmarked via `pacerun.sh` and `paceresults/`
- Used to evaluate **heuristic solvers** (KMA, DKMA, GNN-KMA v1/v2/v3) against real competition-grade instances
- PACE winner results in `pace2022_winner.csv` used as comparison baseline

Why PACE 2022? Our synthetic dataset covers diverse graph families but may not capture the nuances of competition-curated real-world instances. PACE 2022 instances serve as a *held-out real-world stress test* for heuristic solvers, enabling direct comparison with state-of-the-art PACE competitors [8].

End-to-End System Architecture



Evaluation Framework

Benchmark Output Format

Each benchmark run produces a CSV with:

- File identity & graph metadata
- Algorithm / family / track tags
- `fvs_size`: solution size
- `runtime`: wall-clock seconds
- `valid`: acyclicity-verified boolean
- `status`: completed / timeout / error

Benchmark Tracks

exact_track: compared against IC/BST optimal
heuristic_track: compared against MA baseline
PACE 2022: compared against official winner

Algorithm Comparison Matrix

Algorithm	Exact	Heuristic
BST	✓	–
IC	✓	–
MA	–	✓
KMA	–	✓
DKMA	–	✓
GNN-KMA v1	–	✓
GNN-KMA v2	–	✓
GNN-KMA v3	–	✓
GNN-DKMA	–	✓

Resume semantics: all benchmark pipelines support deterministic restart - interrupted runs continue from last completed file.

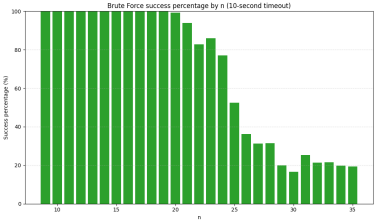
Exact solution Checker

Results

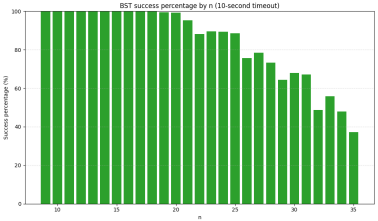
Solver	Mean score (%)	Instances completed Before Timeout
undirected_BST_exact	100.0	9046
undirected_IC_exact	100.0	9906
undirected_brute_force	100.0	7263

- **Instance count:** Number of instances that completed before timeout.
- **Mean score (100%):** For every completed instance, the solver returned an optimal solution among the instances it finished.
- As all three solvers report 100%, the **brute force**, **IC**, and **BST** exact solvers produced the same optimal answers on their completed instances.

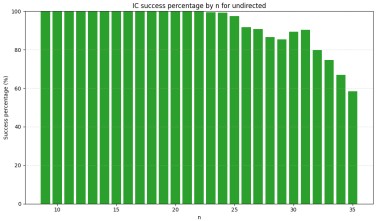
Exact success percentage by graph size



Brute Force exact success rate vs. n

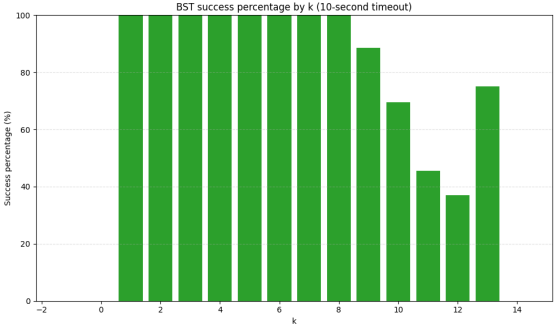


BST exact success rate vs. n

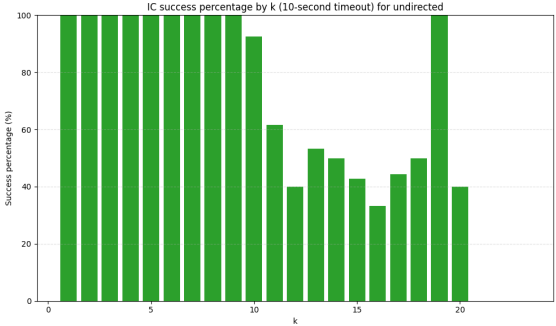


IC exact success rate vs. n

Exact success percentage by solution size

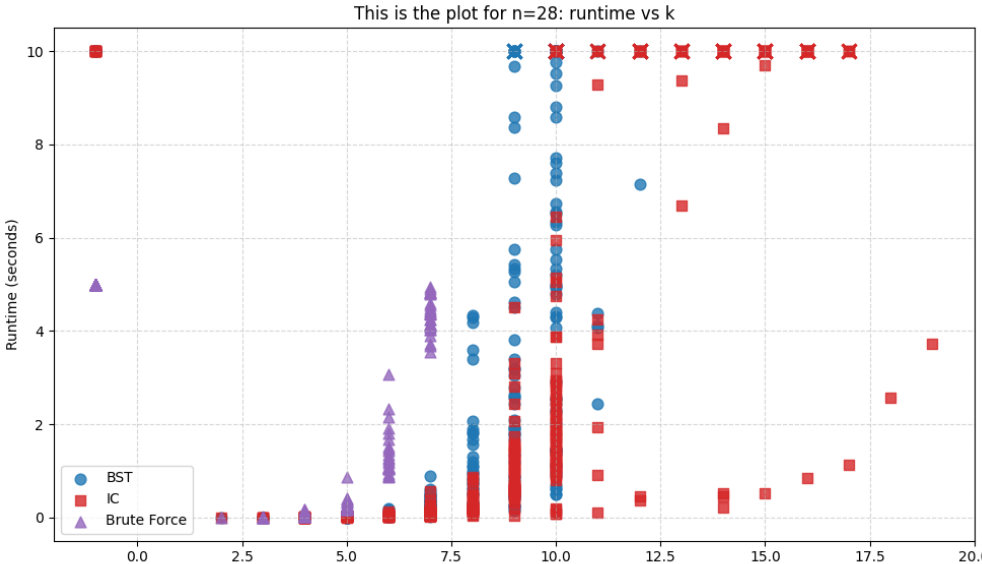


BST exact success rate vs. k

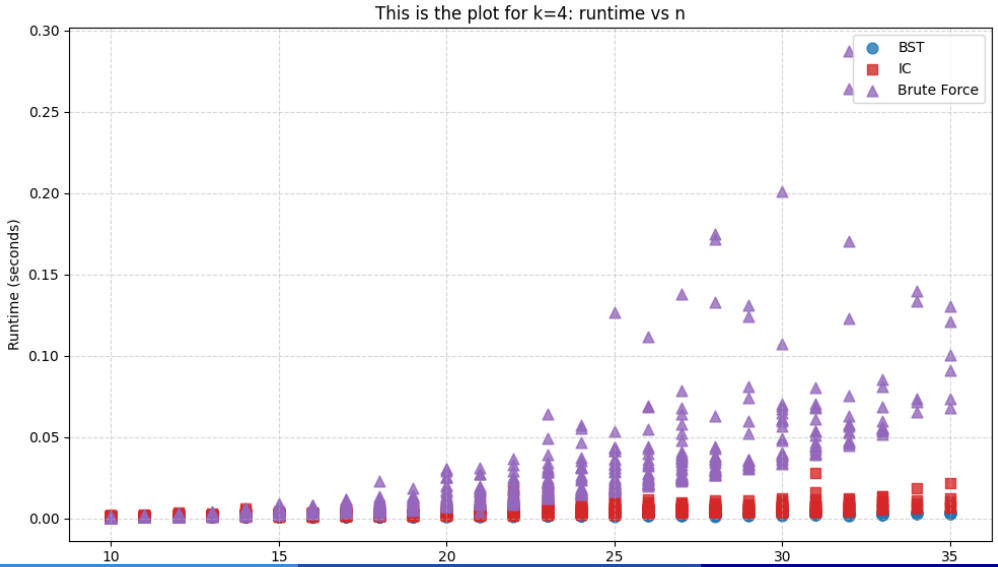


IC exact success rate vs. k

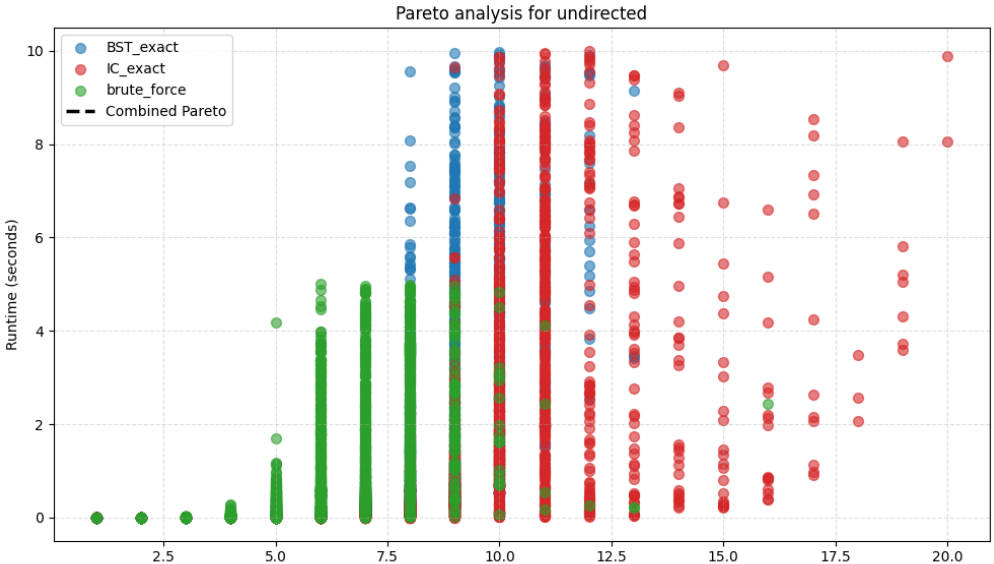
Undirected runtime vs. k for $n = 28$



Undirected runtime vs. n for $k = 4$



Pareto analysis of solver performance



PACE 2022 test scoring

- Each PACE 2022 problem uses a separate 100-point score for each test case.
- If three valid solutions a, b, c have sizes 89, 91, 85 then a's score is $100 \times \frac{85}{89}$ for that case.
- The score formula is $100 \times \frac{\text{OPT FVS}}{\text{solved FVS}}$.
- Total marks are the mean over all test case scores.
- For invalid FVS outputs, we use the fallback size n and score accordingly.

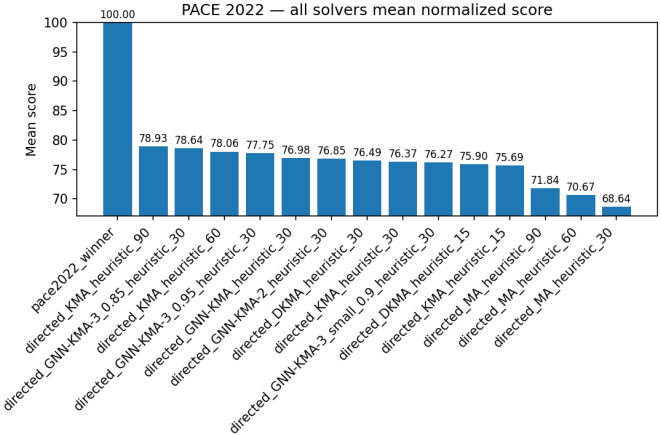
PACE 2022 competitor benchmark

- We benchmarked against the PACE 2022 winner implementation by Sylwester Swat.
- Author: Sylwester Swat, Poznań University Of Technology.
- GitHub: `https://github.com/swacisko/pace-2022`
- His reported score: **99.9117834156**.
- We ran all available heuristics on the PACE 2022 test cases and kept this result as our competitor baseline.
- Because the winner's score is so close to 100, this gives us a realistic reference for actual PACE scoring.

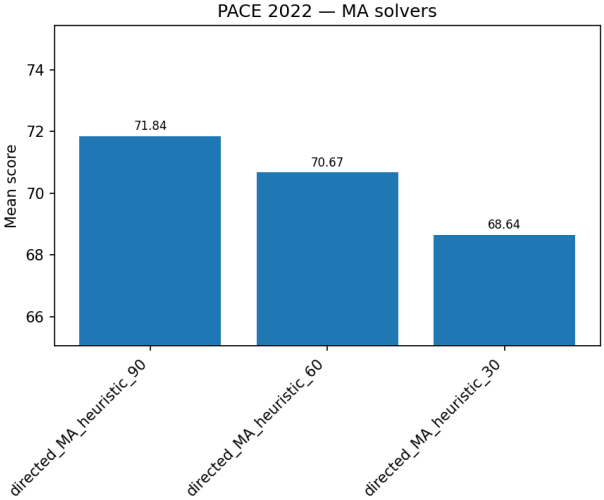
PACE 2022 heuristic results (descending score)

- **pace2022 winner**: 100.0000 (based on 200 instances) - total time=600.00s
- **directed KMA heuristic 90**: 78.9312 (based on 200 instances) - total time=18709.90s
- **directed GNN-KMA-3 0.85 heuristic 30**: 78.6449 (based on 200 instances) - total time=15617.54s
- **directed GNN-KMA-3 0.95 heuristic 30**: 77.7510 (based on 200 instances) - total time=15302.61s
- **directed GNN-KMA heuristic 30**: 76.9814 (based on 200 instances) - total time=8380.73s
- **directed DKMA heuristic 30**: 76.4942 (based on 200 instances) - total time=8367.24s
- **directed KMA heuristic 30**: 76.3657 (based on 200 instances) - total time=7820.26s
- **directed GNN-KMA-3 small 0.9 heuristic 30**: 76.2655 (based on 200 instances) - total time=16480.64s
- **directed DKMA heuristic 15**: 75.8974 (based on 200 instances) - total time=4641.90s
- **directed KMA heuristic 15**: 75.6944 (based on 200 instances) - total time=4209.18s
- **directed MA heuristic 90**: 71.8443 (based on 200 instances) - total time=18281.50s

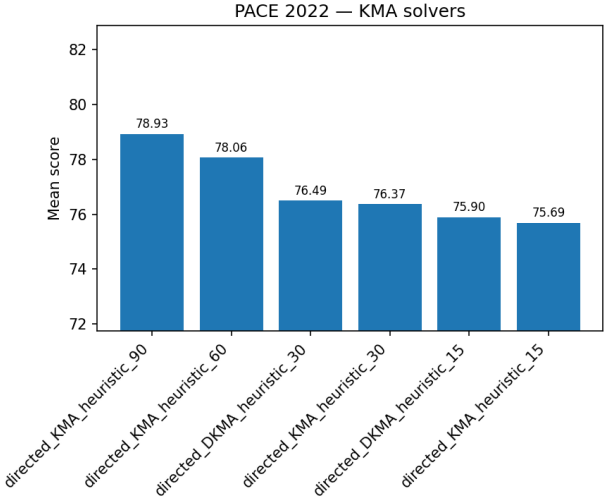
All solvers mean score



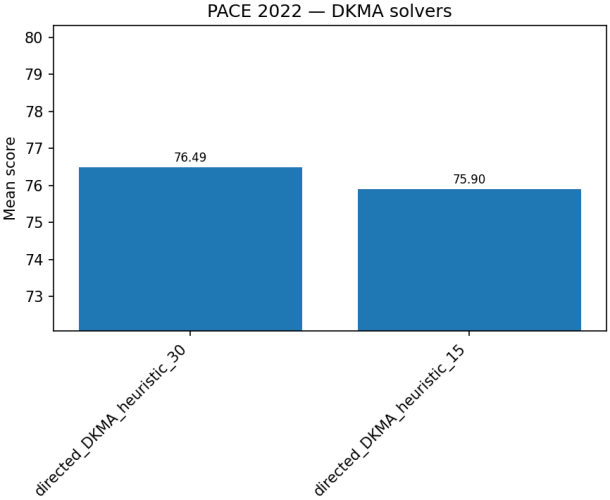
MA models mean score



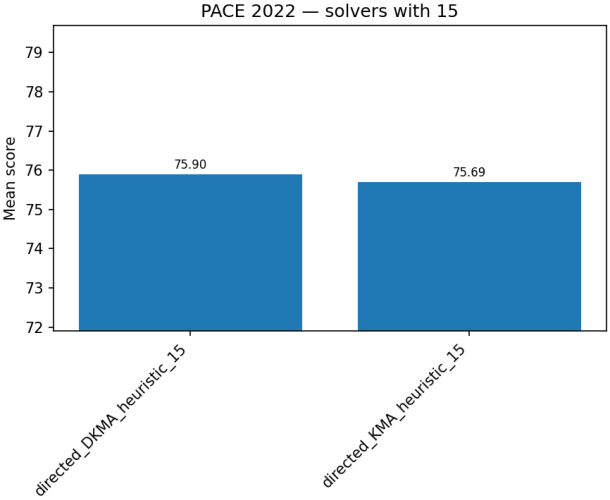
KMA models mean score



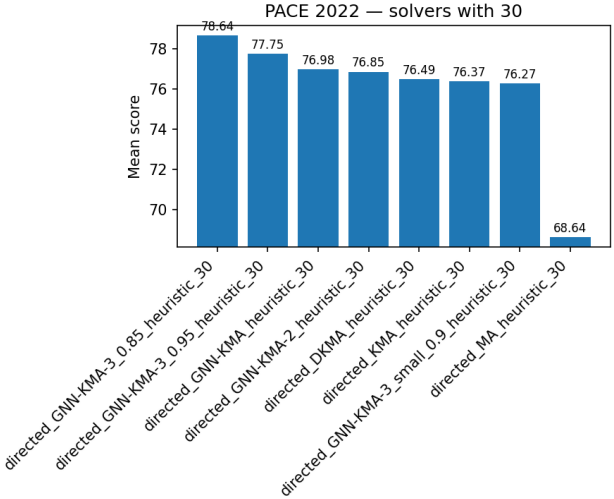
DKMA models mean score



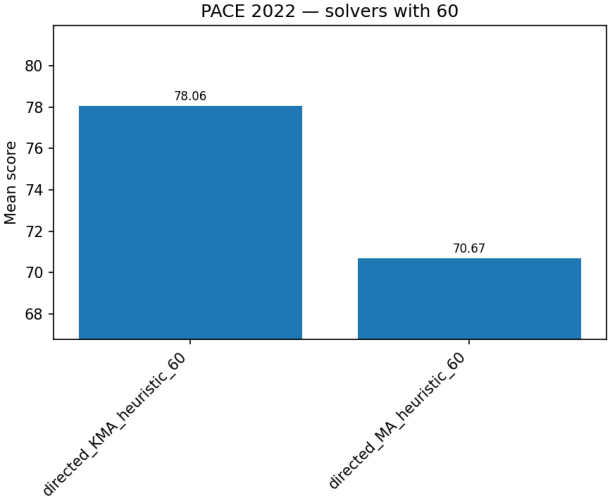
Group 15 mean score



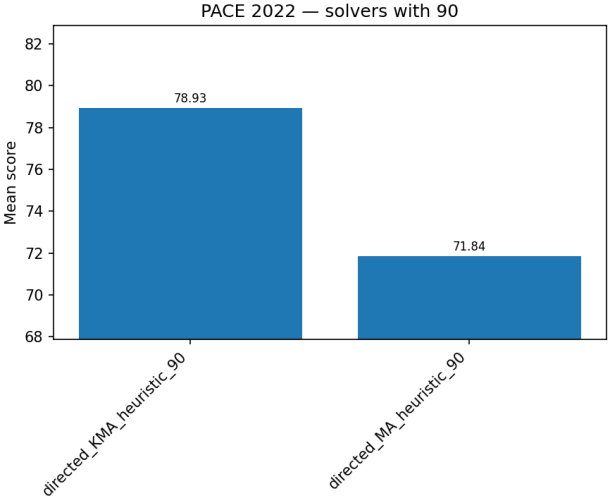
Group 30 mean score



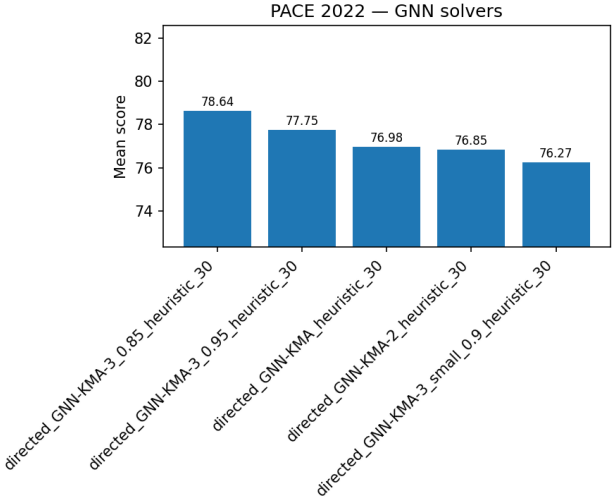
Group 60 mean score



Group 90 mean score



GNN models mean score



Summary & Contributions

Algorithm Contributions

- **KMA**: first-class kernelization (SCC reduction for directed; degree-2 bypass for undirected) integrated with C++ memetic search
- **DKMA**: dynamic re-kernelization loop, SHORTONE contractions [10], topological diversification, gain local search (1-1 / 2-1 swaps)
- **GNN-KMA v1/v2/v3**: progressive feature and architecture improvements (GraphSAGE → GATv2 [12], 3 → 16 channels, NLL → asymmetric precision loss)
- **GNN-DKMA**: GNN guidance + dynamic kernelization

Dataset Contributions

- 100K undirected + 100K directed labelled FVS instances
- 5 graph families, 2 difficulty tracks, real+synthetic mix
- Solver-verified ground-truth labels (IC for exact, KMA for heuristic)
- Publicly released on Kaggle (100K and 20K versions)
- PACE 2022 integration for real-world benchmarking

GitHub:
`https://github.com/TAR2003/Feedback_Vertex_Set_Problem_Algorithm_Project_4-2`

Dataset (100K):
`https://www.kaggle.com/datasets/tawkirazizrahman/fvs-synthetic-dataset-100k`

Dataset (20K):
`https://www.kaggle.com/datasets/tawkirazizrahman/fvs-synthetic-dataset-20k`

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Thank You

Questions & Discussion

https://github.com/TAR2003/Feedback_Vertex_Set_Problem_Algorithm_Project_4-2

https://www.kaggle.com/datasets/tawkirazizrahman/fvs-synthetic-dataset-100k

https://www.kaggle.com/datasets/tawkirazizrahman/fvs-synthetic-dataset-20k